

Sammy Studio Server

Engineering specification v1 for the studio executor

For Utkarsh (UG). From Dan Poore and the Sammy Rane studio. 27 September 2026.

One sentence. Build one service that owns the provider logins, the cast, the rules, the assets and Dan's verdicts, so that an operator model (Claude, Perplexity, anyone) can only make a take by asking the service, and the service can only spend when the request passes every gate, records the result as unreviewed evidence, and leaves a ledger.

Why it is worth building. The studio already makes full-song live videos of an AI band with locked faces: four locked masters in a week (Whisky v16, Shadow stadium v20, Troubadour v5, four teasers). Every dollar it wastes and every hour Dan loses come from the same place: the gates live in one chat client on one Mac, the browser is the API, and the state is markdown. This document turns what works into a machine.

Definition of done for phase 1. From the request "Cass, still 16, driver 22, gap at 1:21", the operator makes three tool calls, no browser is visible to anyone, and the take lands in the store with its contact sheet and a PASS from every machine gate, or a FAIL that names the reason before any credit is spent, in under five minutes.

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1. What the studio makes, and what exists today

The product. Performance videos of Sammy Rane and Westbound, an AI band with eight locked faces (Sammy, Cass, Max, Rex, Tre, Leni, Lizzy, Blaze), singing Dan's Suno masters live in three rooms (the Fillmore, a stadium at night, a stadium in daylight). A video is 20 to 80 shots. A sung shot is a still of Sammy animated by MiniMax H3 with the vocal slice as audio, so the lips are on the words. A band shot is a still of one or two members animated by Motion Sync from a real player's clip retimed to the song. Crowds are real footage carried into the room. Cuts are JSON sheets rendered by ffmpeg.

The constraint. Every generation costs credits (a 9 s take at 2K is about 700 credits; a day can burn 13,000) and the only judge of a face is a read at full size against the locked hero file. Dan's word alone locks anything. The studio's rules (30 numbered, each paid for) are in LOCKED_RULES.md; the method is PLAYBOOK.md; the failure catalogue is HANDOFF.md. Read those three before writing code.

1.1 Components that already work (as scripts in build/)

Script	What it does today	Becomes
spend_gate.py	Checks a frame is approved (sha256 in APPROVED_FRAMES.json), the people are cast, the route is legal (API = Sammy alone), the audio exists, prints the cost, writes a 30-minute single-use token .spend-ok with a count	Authorization service (section 5)
hero_gate.py / checklist_sheet.py	Puts every face in a frame beside its hero file (front and profile) at full size on one sheet; --approve records sha256 + who + when + the read text	Gate service hero_gate (section 7)
face_audit.py	Crops every person at every second of a take beside the hero	Gate service face_audit
xc_check.py	Cross-correlates a take's audio against the master; must print the slice start	Gate service xc_check
retime_driver.py	Retimes a real player's clip to the song's beat grid, phased to the slot's first beat, sidecar JSON	Driver library (section 8)
openart_motion_sync.py	Playwright in the studio's own Chrome profile: uploads, verifies each upload in the DOM, reads the balance, presses Generate once, polls, downloads, writes a contact sheet, logs credits before/after	OpenArt browser adapter (section 6)
oa_pull.py / fillmore_collect.py	Pull finished results by creation id (never 'newest'), name by slice, xc-check, register UNREVIEWED	Landing pipeline (section 8)
cut_gate.py	Refuses a cut whose takes are unregistered, dead or uncropped strangers; builds an edge sheet (first/mid/last, full frame plus left and right quarters) that a human must read before the pass file is written	Render gate (section 10)
cut.py / check.py / sheet.py	Validates and renders a cut sheet shot by shot (streams, no OOM), integrity check, frame sheet	Render service
hooks/*.py	Five PreToolUse/PostToolUse hooks in Claude Code: generation guard, browser click gate, QA gate on render/show/send, render verify, tab recorder. Fail closed	Replaced by the executor: the hooks become one policy engine in the service
backup_media.sh / push_knowledge.sh	Drive mirror of media; GitHub mirror of know-how and heroes	Backup job + the same repo

1.2 Registers that already exist (keep the shape)

- **HEROES_LOCKED/**: the eight faces plus profile files, read-only, SHA256SUMS. Sources of truth in four places with the same bytes (studio, Drive root, GitHub, and one OpenArt Character per person).
- **handoff/<piece>/frames/APPROVED_FRAMES.json**: sha256, approved time, by, the per-person read text, dan_eye. A frame not here with a pass cannot be sent to Dan or animated.
- **handoff/<piece>/TAKES_REGISTRY.json** and **handoff/library/<room>/TAKES_REGISTRY.json**: people, crowd, strangers, dead, crop_ok, why. A take not here cannot enter a cut.

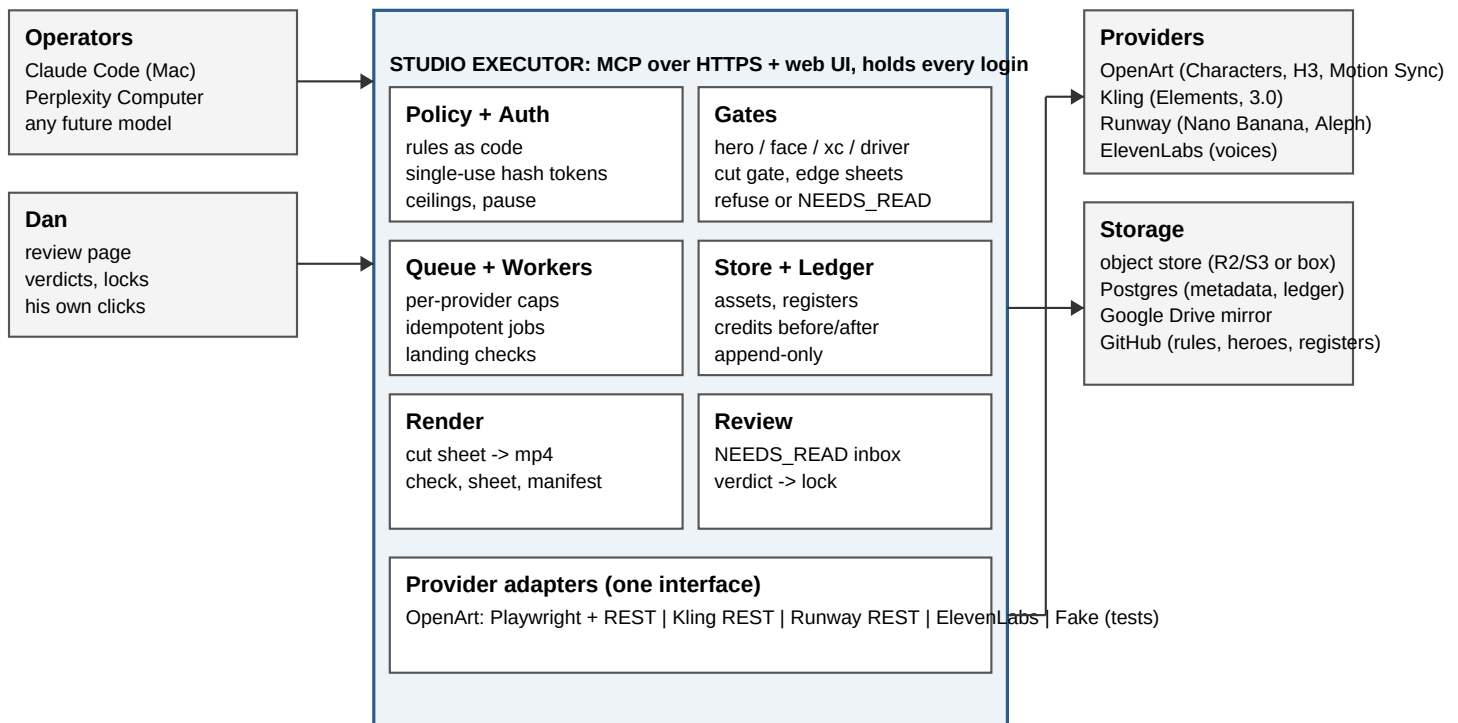
- **cuts/*.json**: a cut IS its sheet (total, frame, shots with start/end/path/src_start/zoom/dim, audio layers). Deterministic render.
- **.cut-gate-ok/<export>.ok**: sha256 of the sheet at the moment it passed. Render, show and send check it.
- **DAN_NOTES.md** per piece: dated log, Dan's words verbatim, credits.

2. The problem, as engineering

Failure class (each happened more than once, dates in HANDOFF.md)	Root cause	What the service must make impossible
A frame sent to Dan or animated with a face nobody read; a take registered 'face-checked' from a thumbnail	The check was a step the operator chose to run	A result is UNREVIEWED until a read with text exists; sending, animating and cutting refuse without it
Generation with no unlock; ~30 Runway/OpenArt calls in a day with no gate	Gates were client hooks; other paths (MCP servers, browser, other operators) had none	The service is the only holder of provider logins; no token, no call
Wrong route: band faces through an API that cannot attach Characters; Runway for a face	Route rules lived in prose	Policy engine encodes rule 17/20: band faces only via OpenArt Characters; API only Sammy-alone or nobody
Double fire (328 credits), stale form after reload, click on Generate meant for the picker (450 credits), upload silently dropped, wrong driver ran	Browser driven by coordinates and hope	Adapter asserts state in the DOM before Generate; idempotency key per request; diff of creation ids identifies the output
13,085 credits overnight before Dan saw a frame	'Go' read as 'spend unseen'	Batch ceiling (two-person proof, then five), per-piece and per-day credit ceilings, Dan's eye on the first result of any new recipe
Strangers, three-armed drummers, second drum kits, folded-arms crowd, mixed rooms, day cut with night	Checks were of the thing last complained about, not the whole frame	Gates run on every person and the whole frame; venue/room and day/night are fields, and a cut with two values is refused
A take reused across slots out of source order; a sung take re-placed at another song time	Cut generator had no rule	Cut gate: sung takes only at their slice time; a take reused across slots runs forward in time
Another operator (Perplexity's Computer) produced and shared a 'master' outside every gate	No shared executor	Only Dan's authenticated verdict sets locked; masters carry a manifest and checksum; a chat attachment is never a master
The Drive backup ran under bash and did nothing, exit 0	No verification of the destination	Backup job verifies checksums at the destination

The two facts that shape the design. First, the identity-holding features (OpenArt Characters, Motion Sync) exist only on the OpenArt website today; the REST/MCP surface covers image and video generation with element references but not the Character editor or Motion Sync. So phase 1 owns a real browser session and drives it deterministically, behind the same tool names that a pure API adapter will satisfy the day the API catches up. Second, no machine can pass a face. Similarity scores are a pre-screen that may refuse; a pass is a recorded human read. The service enforces that the read exists; it does not pretend to be it.

3. Architecture and the trust boundary



Operators never touch a provider or a register directly. Every arrow into a provider carries a consumed token and lands as UNREVIEWED evidence.

Trust boundary. Provider credentials, the spend tokens, the registers and the ledger live inside the executor. Operators hold an MCP bearer token with a role. They can propose, ask for a token, submit a job with that token, read evidence, and write a read (text) on an item. They cannot write a token, mark a pass, set locked, delete, or reach a provider. Dan holds the owner role: verdicts and locks. The service is deployable on the Mac in the house first (it needs the Chrome profile that is logged into OpenArt) and later on a small VM with the browser session moved there.

Language and shape. Python 3.11+, FastAPI (HTTP + MCP tools over Streamable HTTP), Postgres, Playwright (Chromium, persistent profile), ffmpeg via imageio-ffmpeg, object storage via boto3 (R2) or a local path in phase 1, a job queue on Postgres (SKIP LOCKED) to avoid a second datastore; Redis only if throughput demands it. The existing build/ scripts are imported as library functions, not shelled out to, once each has a thin function signature.

4. Data model

Everything below is a table (Postgres) with a JSONB column for provider-specific detail. Files live in the object store under a content-addressed path (sha256) with a human alias. Nothing is ever deleted; status changes are appended to an events table.

Entity	Key fields	Invariants
person	id, name, locked_since, attributes (text per attribute: hair, beard, cap, glasses, sleeves, tattoos, instrument, wardrobe variants), hero_files[] {kind: front profile fullbody, sha256, path}, provider_ids {openart_character, kling_element}	Eight rows. Hero files immutable; a new hero is a new row version with Dan's verdict id. Attribute text is a reminder; the file is the judge
room	id, name, kind (fillmore stadium_night stadium_day), plate_sha256, keeper_plates[]	A cut may reference exactly one room kind
song	id, master_sha256, duration, bpm, vocal_map (phrases[] start/end), beats[], lyrics_timed[]	vocal_map built once from the Demucs stem at its own sample rate; assert stem length = master length
slice	id, song_id, start, end, kind (vocal_only full_mix none), sha256	Named by song start; a sung take is welded to its slice
frame	id, piece_id, sha256, path, room_id, people[], source (mj openart runway edit_of:frame_id), status (UNREVIEWED PASS HELD DEAD), reads[] {who, when, per_person_text, sheet_sha}, dan_eye {when, words}	status PASS requires at least one read with text per person; DEAD is terminal
take	id, piece_id, frame_id, job_id, provider, model, settings, slice_id, people[], crowd, strangers, crop_ok {by, ax, ay}, dead, xc {offset, similarity}, face_audit_sha, status (UNREVIEWED PASS DEAD), reads[]	A take enters a cut only with status PASS; a sung take only at its slice start
driver	id, source (file, licence), instrument, framing, tempo, retime {factor, phase}, body_visible, instrument_height	driver_fit gate compares to the subject still before spend
cut	id, piece_id, sheet (JSON), sheet_sha256, room_kind, gate_pass {sha, when, edges_sheet_sha, looked_by}	Render refuses if sheet_sha != gate_pass.sha
export	id, cut_id, file_sha256, check_report, verdicts[], locked {by, when, words}, manifest	locked only via a verdict by the owner role
authorization (token)	id, inputs_hash, inputs (canonical JSON), count_total, count_left, cost_ceiling, issued_to, issued_at, expires_at, consumed[] {job_id, at}	Single use per count; consumed only by the executor; never returned to the operator in writable form
job	id, token_id, provider, request (canonical), idempotency_key, status (QUEUED RUNNING LANDED FAILED REFUNDED), provider_job_id, credits_before, credits_after, result_asset_ids[], error	One job per count; no automatic retry; a FAILED job releases nothing (the count is spent unless the provider refunded and the balance proves it)
verdict	id, subject (frame take cut export), subject_id, by (person), role, words (verbatim), timecode, decision (approve reject lock dead note)	Append-only. Dan's words are stored verbatim and shown next to the subject
ledger	id, at, actor, action, subject, credits_delta, balance_after, token_id, reason	Every allow and every refusal is a row

5. Authorization: single-use, hash-bound spend tokens

This is the heart. Today the token is a file that any process on the Mac can write. In the service it is a row that only the policy engine can create and only the job runner can consume.

5.1 Flow

```
operator  -> POST /authorize {piece, shot, mode: frame|video, frame_id, people[], room_id, provider, model,
                             settings{res, seconds, aspect}, slice_id, driver_id, count, what, dan_words_ref}
executor  -> canonicalize -> inputs_hash = sha256(canonical JSON)
           -> policy.check(inputs)          # section 5.2; any FAIL returns reasons, no token
           -> cost = pricing[provider][model](settings) * count ; refuse if > ceilings (piece/day/batch)
           -> token = insert authorization(count_left=count, expires_at=now+30min, cost_ceiling)
operator  -> {token_id, cost, references_to_attach[], warnings[]} # the operator says the cost out loud
executor  -> POST /jobs {token_id, request} # request must hash to the token's inputs_hash (+ count index)
           -> verify hash, expiry, count_left>0, role ; count_left -= 1 (atomic) ; enqueue job
           -> worker: adapter.submit(request, idempotency_key=job.id) ; record credits_before/after
           -> landing: fetch asset -> store -> run landing gates -> status UNREVIEWED -> notify
```

5.2 Policy checks (each returns PASS or a named FAIL; all must pass)

Check	Rule it encodes	Data it needs
unlock	Rule 1: generation only after Dan's word, recorded verbatim with a time, valid 12 h	dan_words_ref -> verdict row of kind 'unlock' for the piece
cast	Only the eight people exist; nobody else in people[]	person table
frame_pass	Rule 18 / spend checklist 4: no video from a frame without status PASS and a read per visible person	frame.reads[]
route	Rule 20: API video only if people == [sammy] or []; anyone else -> provider=openart_web with Characters attached. Rule 17b: provider runway never with a person whose face must hold	people[], provider
characters_max	Two Characters per OpenArt frame; more -> refuse with the swap-chain instruction	people[] for mode frame
profile_hero	Rule 17b(c): a profile or nose-to-nose two-shot requires a locked profile hero for each person in it	person.hero_files kind=profile; request.pose
rex_background	Rule 29: Rex only inside wides; refuse a solo or close-up of Rex	people[], shot size
crowd_band	Rules 22/24: never a crowd and the band in one generation; crowd wide = real footage carried in (Aleph), not generated	request.crowd flag, people[]
room_continuity	One room kind per piece cut; day and night never mix; a frame's room must match the piece's room	frame.room_id, piece.room_kind
slice	Sung take needs a vocal slice named by song start and <= provider's audio cap (15 s); non-singing Sammy shots must not carry a slice with vocal	slice.kind, song.vocal_map
driver_fit	Motion Sync driver: body visible and lit, instrument at the subject's height, retimed to the song	driver row, subject frame
ceilings	Per-piece, per-day and per-batch credit ceilings; a new recipe or room gets ONE result shown to Dan before more (two-person proof, then five)	ledger, piece config
pause	A global pause set by the owner blocks new jobs	config

Determinism. The canonical JSON sorts keys, lowercases names, resolves every file to its sha256 and every setting to its enum. The same request always hashes the same; a request that differs by one pixel of the frame is a different token. That is what stops the 'I changed the frame after the gate' class of error.

6. Provider adapters

One Python protocol, four implementations plus a fake. The tool names never change when a provider moves from browser to API.

```
class Provider(Protocol):
    name: str
    def balance(self) -> int
    def generate_image(self, req: ImageRequest, idem: str) -> ProviderJob
    def generate_video(self, req: VideoRequest, idem: str) -> ProviderJob      # element2video: frame + audio + chara
    def motion_sync(self, req: MotionSyncRequest, idem: str) -> ProviderJob    # subject still + driver clip
    def edit_video(self, req: EditRequest, idem: str) -> ProviderJob           # Aleph-class: room swap, crowd carry
    def status(self, job: ProviderJob) -> JobStatus                          # PENDING RUNNING COMPLETED FAILED REF
    def fetch(self, job: ProviderJob) -> list[Asset]                         # bytes + width/height/duration/fps/ha
    def list_recent(self, since) -> list[ProviderJob]                       # the truth for 'did it fire'
```

6.1 OpenArt, browser adapter (phase 1, the one that holds identity)

Playwright with a persistent Chromium profile that Dan signs into once (the studio already runs this as .pw-chrome). Everything the 26 Sep sessions learned by hand becomes assertions in code:

- **State assertion before Generate:** model name on the page, references counter equals 1 (plate) + images + Characters ticked, every @chip present in the prompt text, output row string (aspect, resolution, seconds), duration slider value read from the Output row text (the first slider in the DOM is the gallery zoom), balance read after it stops animating. Any mismatch raises before the click.
- **Characters dialog:** tick tiles one at a time, wait, re-read the tile labels (the library re-orders; a tile from another session can appear); a typed @NAME that resolves to no ticked Character is dead text.
- **Uploads:** through the references dialog's own input; verify the file appears at its own pixel size (a subject reverting to the hero's portrait aspect is a known trap); JPEG q95 <= 10 MB; no hard-linked files; videos only through History > Uploads and confirmed by the page's video src.
- **Audio:** the page caps attached audio at 15 s total; remove the old chip before the next slice.
- **Fire once:** snapshot creation ids before the click, click Generate by element handle, then poll list_recent for the new id. Never trust the Generating flag or the balance text (both lag). Idempotency: if the diff already shows the request's fingerprint (prompt hash in the record), do not click again.
- **Land:** download by creation id from the CDN url in the record (public HTTPS, curl works), never 'newest'. Two jobs finish out of order on one account, and other operators share the account.
- **Cost:** read the Generate button price before the click; store credits_before/after; treat a refund (error 1002 on a bad driver) as REFUNDED only when the balance proves it.
- **Concurrency:** OpenArt allows 8 parallel generations per account; the worker pool caps at 6 to leave room for Dan's own clicks.

6.2 OpenArt, REST adapter (phase 1b; cheaper, no browser, no Characters editor)

- Read-only calls work today: account, model list and forms, model cost, project list, creation list and get, upload list.
- fal-h3-max and minimax-h3 element2video take visualReferences[] of type image, audio and element (the Character ids) in one array. That is exactly the website's shot with a 10 percent discount. Use it for every sung take once the frame exists.
- upload_sign returns a PUT url; the studio's CLI call failed because size was sent as a string. Send integers. Then upload_metadata_get confirms the visualReference.
- The Suite's 'Auto' image router has no API id; Nano Banana Pro (40 credits, up to 14 refs, 4K) is the stand-in. Confirm with one side-by-side.

- What has no API: creating or editing a Character, Motion Sync. Those stay on the browser adapter until they appear.

6.3 Kling (REST/MCP, Dan's Ultra plan)

- Use for: solo band members and Sammy not singing (3.0 / 3.0 Turbo, 10 credits per second at 1080p on Turbo), still edits (image_to_image omni: one Element per pass, multi-reference for wides), prop swaps in place (reliable), moving a prop (not reliable).
- Upload: file ticket then a multipart POST; tickets expire in ~10 minutes.
- Known limits: cannot hold Sammy's face in a swap (loud features hold, subtle ones do not); multi-character binding fails; native audio off for anything that will be cut to a master.
- Kling 3.0 motion control is what OpenArt's Motion Sync wraps; if Kling exposes it directly, the motion_sync method gets a second implementation.

6.4 Runway (REST)

- Use for: one-attribute edits on non-face things (20 credits per 2K Nano Banana image), crowd plates and room carries with Aleph (168 credits per 6 s), candidate frames from a take. Policy refuses any request where a person's face must hold.
- The Runway inline viewer in a chat client shows every result, including rejects, before a gate runs. The service removes that by never returning a result to the operator before landing gates have run.

6.5 ElevenLabs

Voices for teasers and spoken lines only (cast IDs in LOCKED_RULES.md; Sammy clone stability 0.4, similarity 0.85, style 0). Never for singing. Multilingual v2 reads stage directions aloud: strip parentheses before synthesis and whisper-check the take.

7. Gates as services and the evidence rule

The evidence rule. A machine gate may refuse. It may produce evidence (a sheet, a number). It may never pass a face. A pass on a frame or take is a read: a person (or a model acting as reviewer, named as such) looks at the evidence at full size and writes what they saw per person. The service stores the read text, the reviewer, the time and the sha256 of the sheet they read. Sending, animating and cutting all require that row. This keeps the honest limit of today's tools (rule 18b: a face you cannot verify is a stranger) while making the read unskippable.

Gate	Input	Output	Refuses when
hero_gate	frame, people[], optional head boxes	sheet: hero front + profile beside each crop, at full size; per-person similarity pre-screen (face embedding vs hero) as a number only	a person is missing from the frame, a head box is empty, or similarity is below a hard floor (refuse, never pass)
checklist_sheet	frame, people[]	full frame + left/centre/right thirds at 2x + hero panels	never; it is evidence for the read
frame_whole	frame	detected instruments, kits, mic stands, people count	an empty second kit, a spare guitar, a stand with nobody, a person count that differs from people[] (ghost-player prevention)
face_audit	take, people[], boxes, step	per-second crops beside heroes, first/middle/last at full size	a person leaves the frame is fine; a similarity collapse mid-take flags NEEDS_READ with the second marked
xc_check	take, master, slice	offset in seconds, similarity	offset differs from slice start by > 60 ms, or similarity < 0.2 against master AND slice (the model invented its own audio)
sync_bobbing	take with several people or a crowd	Pearson r of per-region motion energy	r above threshold between regions (everyone moving in step, rule 22)
driver_fit	driver clip, subject still	body visibility, brightness, instrument height match, tempo vs song	body dark or hidden, instrument at the wrong height, tempo not retimed
cut_gate	cut sheet, registers	edge sheet (first/mid/last, full + left/right quarters at the cut's zoom) for every crowd/band shot	unregistered take, dead take, stranger without the registered crop, sung take not at its slice time, reused take out of source order, two room kinds, day + night, a black slug, a still

Every gate returns {status: PASS | FAIL | NEEDS_READ, reasons[], evidence[] (asset ids)} and writes a ledger row. Gates are pure functions over stored assets so they can be re-run when a rule changes and the results diffed.

8. Job runner, workflow and landing

- **Queue:** jobs table with status and a worker lease (SELECT ... FOR UPDATE SKIP LOCKED). One worker pool per provider with its own concurrency cap and rate limit. A job carries its token id; the worker re-verifies the token was consumed for this job before submitting.
- **No automatic retry.** A FAILED job stays FAILED with the provider's error. The operator may ask for a new token. A REFUNDED status is set only after balance() proves the refund.
- **Timeouts:** image 5 min, H3 768p 5 min, H3 2K 20 min, Motion Sync 10 min; past the timeout the job is marked STALE and list_recent is consulted before anything else happens.
- **Landing pipeline** (automatic, zero credits): fetch by id, store with sha256, probe (width, height, duration, fps, hasAudio), then xc_check (if a slice), face_audit sheet, sync_bobbing (if several people or crowd), contact sheet at 2 fps, then status UNREVIEWED and a notification to the operator and the review inbox. Nothing is returned to the operator before this runs.

- **Driver library:** intake of Dan's phone clips and licensed stock (licence recorded), tags (instrument, framing, tempo, instrument height, body visible), `make_driver(clip, song, start, strokes_per_beat, crop)` runs beat tracking, retime, phase and crop and stores the beat anchor. Drivers that failed (dark drummer, seated acoustic at chest height) are kept with their failure reason so they are never picked again.
- **Batches:** a batch is a list of tokens under one ceiling; the policy allows a first batch of two for a new recipe or room, then five, each accepted by a verdict before the next.

9. Asset store, registers, ledger, locks and backup

- **Layout:** objects/<sha256[:2]>/<sha256> with the original filename as metadata; aliases in the DB (piece, shot, take name). Heroes and *_APPROVED exports are immutable objects with a lock flag the API refuses to clear.
- **Registers:** the service is the writer of APPROVED_FRAMES.json, TAKES_REGISTRY.json and cuts/*.json; it exports them to the git repo on every change so the Mac scripts, Perplexity and a human with a text editor keep working from the same files during the transition. Import on startup reconciles the repo's files into the DB (sha256 wins).
- **Ledger:** append-only; every authorize, refuse, submit, land, read, verdict, lock, backup with actor, credits delta and balance after. A daily summary is posted to the same webhook the ops watchdog uses.
- **Tags at intake are mandatory:** a take without people[], room, song and slice cannot be stored. This is where 'just build it' projects die; refuse the write instead.
- **Backup:** nightly and after every lock, rsync to the Drive mirror plus a checksum manifest compared at the destination; the job fails loudly on a mismatch (the 26 Sep silent-bash failure is the fixture).
- **Resolution floor:** on approval of a Motion Sync take (1280x720) run one upscale to 2K and store it as the master take; derivatives come from that.

10. Review loop and render service

10.1 Review page

- An inbox of NEEDS_READ and UNREVIEWED items, newest first, each with its evidence (checklist sheet, face audit, edge sheet, the playing clip with the master under it at its song time).
- A read form per person: the reviewer writes what they saw (hair, beard, cap, glasses, sleeves, tattoos, instrument). Submitting a read with an empty field is refused. The read is what promotes UNREVIEWED to PASS.
- Verdict controls for the owner role only: approve, reject with words and timecode, dead, lock. Dan's words are stored verbatim and shown beside the item forever (today they are quoted in DAN_NOTES; the DB keeps that habit).
- Full-screen playback on the Mac remains a one-click action (show_fullscreen.sh) for exports that carry a gate pass.

10.2 Render

Provider note (27 Sep 2026). DaVinci Resolve 21.1 Studio ships a native MCP server (88 tools: project, media pool, timeline, colour, Fairlight, render queue, export; File > Setup AI Assistants detects Claude Code). It is the natural render, grade and sound engine behind this section: the cut sheet stays the record and cut_gate stays in front, Resolve builds the timeline, applies the one-LUT grade, mixes the crowd bed, and renders presets and 9:16 reframes. Its AI B-roll is never used. The executor adds Resolve's render and export tools to the same pass check as build/cut.py.

- render(cut_id): refuses unless cut.sheet_sha256 == cut.gate_pass.sha; streams frames shot by shot to the encoder (the whole-song render used to OOM); runs check.py (size, 24 fps, duration, audio present, no clipping, not silent); writes the frame sheet and a manifest {cut sha, take ids and shas, gates passed, renderer version}.
- excerpt(cut_id, from, to) for a fast look at a stretch.
- Cut sheet grammar enforced at validate time: sung takes at their slice time with src_start 0; a take reused across slots in forward source order; nothing over 5 s unless the operator marks it a held close-up; no black slugs, no stills; one room kind; a sound layer under every cut.

11. Security, accounts, ceilings, pause

- Provider secrets in a vault (1Password CLI or SOPS-encrypted file) read by the executor at start; never in the repo, never returned by any tool. Vendor accounts in the company's name with 2FA.
- MCP bearer tokens per operator with roles: operator (propose, authorize-request, submit, read evidence, write reads), reviewer (reads and non-locking verdicts), owner (Dan: unlock words, verdicts, lock, ceilings, pause). Tokens rotate; every call is logged with its token id.
- Ceilings: per piece, per day, per batch, per single job; defaults from this month's real numbers (a 4-minute song at list price is about 10,500 credits; a careless day was 13,000). Over a ceiling, the answer is a refusal with the number, not a prompt.
- Pause: one switch; while set, /authorize returns PAUSED and workers stop taking jobs.
- The browser profile is a secret too: the machine that runs it is the executor host; nothing else uses that profile.
- Rights: record the licence of every driver clip and the vendor terms for outputs at intake; the store refuses a driver without a licence field.

12. Observability and the numbers that route work

Metric	Why it matters	Source
Credits per approved second, per provider and per person	The only cost number that matters; routes work to the engine that wins for that person	ledger + verdicts
First-try approval rate per recipe	Tells which recipes are locked in and which still burn	verdicts
Gate refusals by reason	A rising reason is a rule the operators keep tripping; a falling one is a rule that can relax	ledger
Sync error in ms, similarity distribution	Catches the model inventing its own audio before a read	xc_check
Drift catches per take (face_audit flags) and strangers found at read time	Measures whether the read is doing its job; a stranger found by Dan is a failed gate	face_audit, verdicts
Time from request to landed evidence	Phase 1 done means under five minutes	jobs
Balance deltas not attributable to a job	Another operator or Dan's own clicks; keeps the ledger honest	balance() polling

Every rejection carries a reason; a repeated reason becomes a check with the date and the take that taught it. Every approval stores the exact recipe (still, driver, prompt, model, settings, offsets) so that 'do Cass like the Troubadour' is a lookup, not a memory.

13. Test plan

13.1 Fake provider

A Fake adapter that returns canned assets (a 6 s colour-bar clip with a known audio slice muxed in, a still with known faces) and records every call. All policy and workflow tests run against it. No test ever spends a credit.

13.2 Denial tests (each must refuse with the named reason)

- no unlock words for the piece; unlock older than 12 h
- token expired (31 min); token count exhausted; request hash differs from the token's inputs (one pixel changed in the frame)
- video request with people [sammy, cass] via provider openart_api or kling (rule 20); any face request via runway (17b)
- three Characters in one OpenArt frame; a profile two-shot of Rex (no profile hero); a Rex close-up (29)
- crowd flag with people[] non-empty (22/24); a frame in stadium_day inside a stadium_night piece; a cut with two room kinds
- a cut with an unregistered take; a dead take; a stranger take at zoom below the registered crop; a sung take placed at the wrong song time; a reused take out of source order; a black slug; a still
- a send of a frame whose status is UNREVIEWED or HELD; a render whose sheet sha differs from the pass
- a lock attempted by an operator or reviewer role; a delete of a hero; a write to a token by an operator
- per-day ceiling reached; batch of five before a two-person proof exists; pause set

13.3 Replayed failures (fixtures from the studio's own history)

Fixture	Date	Expected outcome in the service
Two-shot registered 'face-checked at 4 s' from a thumbnail; Max drifted in the first second	26 Sep	Landing face_audit flags second 0-1; status NEEDS_READ; cut_gate refuses until a read exists; the read form demands text for Max
Side wide passed on a blurred profile crop; forearm tattoo in the full frame	26 Sep	checkboxlist_sheet is the evidence; a read that says 'locs, grey tee' with no tattoo field is refused
Full-stage take with three stranger silhouettes cut in after rule 27	26 Sep	take.strangers=true with no crop_ok -> cut_gate FAIL
Seven Characters in one frame; 3 of 7 strangers	26 Sep	characters_max refuses at authorize; suggests the swap chain
13,085 credits overnight on a new venue unseen	26 Sep	ceilings + two-person proof refuse the third job until a verdict exists
Double fire after the Generating flag cleared (328 credits)	21 Sep	idempotency key + creation-id diff: second submit is a no-op
Motion Sync ran on the un-retimed driver; upload silently dropped	25 Sep	adapter upload verification fails the job before Generate
MiniMax invented its own audio on the 82.55 slice twice	26 Sep	xc_check similarity < 0.2 against master and slice -> FAIL at landing, no read requested
Hero copied from 'newest file in Downloads' was a Cass frame	26 Sep	hero writes require a named source file and Dan's verdict; there is no 'newest'
Drive backup under bash, exit 0, nothing copied	26 Sep	backup job compares checksums at the destination and fails loudly

13.4 Contract tests against real providers

One tiny call per adapter method under a 300-credit ceiling, run by hand before a release: an image at the cheapest size, a 5 s 480p video, a balance read, a list_recent, an upload. Recorded responses become the fake's fixtures.

14. Delivery plan

Milestone	Scope	Acceptance
M1 (week 1-2): the executor core	FastAPI + MCP tools; Postgres schema (section 4); policy engine with every check in 5.2; token issue/consume; ledger; Fake provider; import of the repo's registers and heroes	All denial tests pass; a fake job goes request -> token -> job -> landed -> UNREVIEWED -> read -> PASS with a full ledger
M2 (week 2-3): OpenArt browser adapter	Playwright profile, image_with_characters, h3_take, motion_sync with every DOM assertion in 6.1; landing pipeline with xc_check, face_audit, sheets	Definition of done: 'Cass, still 16, driver 22, gap at 1:21' lands with evidence in under five minutes, no visible browser, cost said before the click
M3 (week 3-4): gates, review, render	hero_gate/checklist/frame_whole/sync_bobbing/driver_fit as services; review page with read forms and owner verdicts; render service with cut_gate and manifests; exports to the git repo	Dan approves a 30 s cut end to end through the page; render refuses on a changed sheet; a lock sets immutable
M4 (week 4-5): the rest of the providers and the store	OpenArt REST for sung takes (10 percent cheaper), Kling, Runway, ElevenLabs adapters; object store; driver library; backup job with verification; metrics dashboard	Contract tests green; a whole piece produced with the Mac hooks switched off
M3b (with M3): rhythm engine v1	Song grid + energy per beat; driver library with driver_fit; beat_check on landing; zoom envelopes in the renderer; assemble(brief) for one section	The 19 Sep live-feel reference cut reproduced from registers and the grid; every cut point on the beat; a band take ranked by beat_check
Phase 2 (month 2-3)	Take index and search (person, shot size, action, room, song, phrase, beat anchor, embedding); derivatives (2K masters, 9:16 reframes with subject tracking, punch-ins); assembly from a brief; coverage planner; publishing through the social connector	'25 vertical 15 s trailers for the Troubadour song' with zero credits unless coverage shows a hole
Phase 3 (when the library is large)	Own the cast: a LoRA per member from HEROES_LOCKED plus approved stills; Wan Animate / LatentSync on a rented H100; same Provider interface	A band shot from our own weights judged against OpenArt on identical inputs

Order of work inside M2, because it is where the money is: (1) the state-assertion layer and the creation-id diff, (2) motion_sync (already prototyped in build/openart_motion_sync.py), (3) image_with_characters with the one-at-a-time tick, (4) h3_take with the audio chip rule. Each lands as a tool the operator can call, with the fake behind it for tests.

Do not build: a video or lip-sync model, an editor, a chat UI, a scheduler that fires generations on its own. The operator proposes; the service decides and spends; Dan judges.

15. The rhythm engine: what makes the videos dance

Dan, 27 Sep: "build something to make the vids dance." The honest starting point: a sung take is welded to its slice (the lips are on the words and it is never retimed or moved), so the dance does not come from stretching Sammy. It comes from four layers the service can own end to end: the band moving like players on the beat, the crowd moving like people out of phase, the camera moving with the music, and the cut landing on the grid. Today each of these is a hand step; here they become one subsystem with measurable output.

15.1 Ground truth: the song grid

- `song.beats[]` from the master (librosa beat tracking, checked by ear once per song and corrected by hand where the drummer pushes); bar length; downbeats; section map (intro, verse, chorus, bridge, solo, outro) from the timed lyric sheet.
- `song.vocal_map[]` from the Demucs vocal stem at its own sample rate (the 0.919 compression bug is the fixture): sung phrases and gaps to the 50 ms.
- `song.energy[]` per beat (RMS of the full mix and of the drum stem) so the engine knows where the hits are, not just where the beats are.
- Every take, driver and cut stores its beat anchor (the song beat its first frame sits on) so anything can be re-phased anywhere in the song without a re-shoot.

15.2 Band motion that is human and on the beat

- **Driver library** (section 8): real players filmed to the track or licensed stock, tagged by instrument, framing, tempo and instrument height. `make_driver` retimes strokes to the song's grid (one or two strokes per beat, chosen per instrument), phases the first stroke onto the slot's first downbeat, crops out feet and pedals, and stores the beat anchor. `driver_fit` refuses a dark, hidden or wrong-height driver before any credit (the seated-acoustic and dark-drummer fixtures).
- **Motion Sync per member**: one still, one driver, one member. Never one prompt for the whole band (that is the one-rhythm tell). The output carries the driver's beat anchor.
- **beat_check** on landing: motion-energy peaks of the take (frame differencing in the player's region) against the song's beat grid; report the share of peaks within 60 ms of a beat and the phase error. The metric is noisy on hand motion; store it as a number for ranking, never as a pass.
- **Dance level** per take: motion energy relative to the driver, so 'half that' (Dan on the Troubadour Cass shot) is a slider: pick the driver whose energy matches the song, or slow the same driver to one stroke per beat (the slow60 recipe).
- **Rex** stays in wides (rule 29); his drum hits in a wide still land on the grid because the wide's driver is the drum stem's onsets, not the beat grid (attempt 4 lesson).

15.3 Crowd motion that is out of phase

- Real crowd plates carried into the room (Aleph) keep real motion; the service stores each plate's motion-energy profile and picks, for a slot, the plate whose peaks do NOT line up with the band's.
- `sync_bobbing` as a gate on any shot with several people: Pearson r between regions above threshold = refused. This is rule 22 as a number.
- Crowd shots never longer than 2 s in a cut (`validate_cut` enforces).

15.4 Camera that moves with the music (free, in the cut)

- The cut sheet already supports zoom (by, ax, ay) and dim; add **zoom envelopes**: a punch-in of 1.0 to 1.12 that lands on a downbeat, a slow drift across a phrase, a snap back on the next hit. These are pixel operations on the render, zero credits, and they read as a camera operator feeling the song.

- Speed ramps only on non-synced clips (band, crowd, room), phased so the ramp peaks on a hit; sung takes are never retimed (rule from 21 Sep: a take only syncs at the time it was shot to).
- Dissolves and hard cuts are chosen by section energy: hard cuts on the chorus grid, a 1.0 s dissolve into the bridge.

15.5 The cut lands on the grid

- assemble(brief): given the section map, the vocal map and the registers, propose cuts that obey the grammar: a synced close-up carries every sung line (whole lines), band and crowd own the gaps, every cut point snapped to a beat (eighth-note grid, clamped so a synced piece never starts before its slice or runs past its clip), angle and size change at every cut, nothing over 5 s unless it is a held close-up, chorus cuts faster than verse cuts, wides only on hits.
- The generator prints the timeline with the beat each cut lands on and the reason for each slot, then cut_gate runs, then the render. The live-feel test of 19 Sep (synced close-up intercut with a band solo on the beat, band at x1.25) is the reference cut the engine must be able to reproduce from its inputs.
- A/B: render two candidate cuts of one section and show both full screen; Dan's verdict trains the ranking (section 12).

15.6 Measured, not felt

Number	What it tells us	Target
Cut points on the beat (share within 40 ms)	The edit dances	above 95 percent
Band motion peaks within 60 ms of a beat, per member take	The player is on the song	rising per recipe; ranked, not passed
Region-to-region motion correlation in band and crowd shots	Nobody moves in unison	below the rule-22 threshold, else refused
Sung-take offset from its slice	Lips on the words	under 30 ms, always
Average shot length by section	Energy of the edit matches the song	verse 3-5 s, chorus 1.5-3 s, close-ups may hold a whole line

Where this goes in the plan: 15.1, the driver library and beat_check are M2/M3 work (they already exist as scripts). Zoom envelopes and assemble are M3/M4. Phase 3 (own the cast) is what makes motion transfer cheap enough to do for every band slot in every song: Wan Animate on a rented GPU with a LoRA per member, the same Provider interface, judged against Motion Sync on identical inputs.

Appendix A. Tool API (MCP over HTTP; all calls idempotent; ids not paths; long jobs return a job id)

```
# proposing and authorizing
authorize(piece, shot, mode, frame_id, people[], room_id, provider, model, settings, slice_id?, driver_id?, count, wh
  -> {token_id, cost, references[], warnings[]} | {refused: [reason...]}
# spending (requires a token; consumed by the service)
image_with_characters(token_id, prompt, plate_id, character_names[], extra_image_ids[], size) -> job_id
h3_take(token_id, frame_id, slice_id, character_names[], prompt, seconds, resolution) -> job_id
motion_sync(token_id, subject_frame_id, driver_id, prompt) -> job_id
solo_take(token_id, frame_id, prompt, seconds, provider='kling') -> job_id           # Sammy alone / no person
edit_video(token_id, take_id, room_plate_id, prompt, provider='runway') -> job_id   # Aleph-class room/crowd carry
job_status(job_id) -> {status, credits_before, credits_after, assets[]}
# evidence and reads (free)
gate(kind, subject_id, params) -> {status, reasons[], evidence[]}
evidence(subject_id) -> [asset ids: sheets, audits, xc]
write_read(subject_id, per_person: {name: text}, looked_evidence_sha) -> read_id    # promotes UNREVIEWED -> PASS
# registers and cuts
find(kind, person?, song?, room?, section?, status?) -> ids
get(id) -> file + metadata
register_take(take_id, people[], crowd, strangers, crop_ok?, why) -> ok             # requires a read
validate_cut(sheet) -> {errors[], timeline}
cut_gate(cut_id) -> {status, edges_sheet_id}      then write_read(cut_id, ...) then render(cut_id)
render(cut_id) -> job_id ; excerpt(cut_id, from, to) -> job_id
# owner only
unlock(piece, words) ; verdict(subject_id, decision, words, timecode?) ; lock(export_id, words)
set_ceiling(scope, credits) ; pause(on) ; balance(provider)
```

Appendix B. Rules as code (LOCKED_RULES.md number -> check)

Rule	Words (short)	Check / where
1	Generation only on Dan's word, 12 h	unlock (5.2)
6	Execute the script literally; a missing shot is black in a teaser, never a still; no reordering	validate_cut; cut sheets are the only render input
12	Never regenerate an approved take	lock flag on exports and takes; authorize refuses a request whose frame+slice match a locked take
14	Every lesson written the same turn	ledger + DAN_NOTES export
16-18, 18b	Heroes are the only band references; every visible man gated at full size; a face you cannot verify is a stranger	frame_pass, hero_gate, face_audit, write_read required per person
17, 17b, 20	Band faces only via OpenArt Characters on the website; Runway heroes are hints; API only Sammy alone	route
17a	Max's hair is wavy to the jaw; the hero file outranks the checklist text	person.attributes are display text; the hero file is on every sheet
17b(c)	No profile two-shot without a locked profile hero	profile_hero
21, 30	Spend checklist; machine gates on every spend path	authorize + tokens + adapters
22, 23, 24, 25, 25a	Crowds never in sync; never with the band in one generation; no matted silhouettes; person-by-person crowds only behind one man; no folded-arms old man (a STOP)	crowd_band, sync_bobbing, read form crowd fields, DEAD status
26, 27	The close-up look (MJ raw, big face, black room, side key, expression); stranger band never leaks	frame.source and room fields; take.strangers with crop_ok; cut_gate
28	The cut gate before any render/show/send	render refuses without gate_pass matching the sheet
29	Rex background only	rex_background
30(e), 30(f)	Other operators work under the same checklist; hooks are Mac-only	the executor is the gate for everyone
venue	Never mix rooms or day and night	room_continuity
Part N	Job to site: sung = H3; band motion = Motion Sync; solo/non-singing = Kling; frames = MJ with heroes or OpenArt Characters; fixes/crowds/rooms = Runway	route table in policy config

Appendix C. Cost table (list prices observed Sept 2026; the pricing module reads these from config)

Engine	Use	Credits
OpenArt Nano Banana Pro, 16:9 4K	frames with Characters	80 per image (40 via API at 2K-4K)
OpenArt MiniMax H3 Max 768p	sung lines, multi-man video	41 per second (6 s = 246, 14 s = 574); 10 percent off via REST
OpenArt MiniMax H3 2K	sung close-ups at 2K	65-90 per second (9 s about 729-810); about 13 min render
OpenArt Motion Sync (Kling 3.0 motion control)	band motion from a driver	210 per 6 s, 280 per 8 s; 1280x720
OpenArt Seedance 2.5	alternate; look reference only	60/s at 480p, 130/s at 720p, about 200/s at 1080p
Kling 3.0 Turbo 1080p	solo / non-singing	10 per second
Kling image_to_image omni	still edits	2-4 per image
Runway Nano Banana Pro 2K	non-face edits, crowd frames	20 per image
Runway Aleph	room / crowd carry	168 per 6 s
Runway Seedance 2 animate	not for us (one rhythm in wides)	216 per 6 s 720p

Appendix D. Script-to-service map

Today (build/)	Service module	Notes
spend_gate.py	policy/authorize.py	same checks, DB-backed token
hero_gate.py, checklist_sheet.py, face_audit.py	gates/faces.py	pure functions over stored assets; embeddings as pre-screen only
xc_check.py, whisky_vocal_map.py, retime_driver.py, song_plan.py	audio/	song, slice, driver, beat grid
openart_motion_sync.py (+ the 26 Sep website recipe in PLAYBOOK)	providers/openart_web.py	Playwright; assertions in 6.1
oa_pull.py, fillmore_collect.py	workflow/landing.py	fetch by id, name by slice, register UNREVIEWED
cut_gate.py, sheet.py, cut.py, check.py, show_fullscreen.sh	render/	gate, validate, render, verify, show
hooks/*.py	removed	the executor is the gate; keep them on the Mac until M4 acceptance
backup_media.sh, push_knowledge.sh	jobs/backup.py, jobs/export_registers.py	verified checksums; registers exported to git on change

Appendix E. Glossary

Term	Meaning
hero	The locked reference image of one person; the only allowed face; immutable; front, profile, full-body files
Character	OpenArt's saved identity built from a hero file; attached to a frame or take by name; the only method that has held band identity
piece	One video (e.g. Shadow at the Fillmore); a folder with its notes, frames, takes, registers, cuts
room / plate	A venue look (Fillmore, stadium night, stadium day) and its approved empty-stage image used as the first reference of every frame
frame / still	A single image a take is animated from; must PASS a read before any video credit
take	A generated clip; UNREVIEWED on landing; PASS after a read; DEAD when Dan kills it
slice	A cut of the master (vocal-only or full mix) starting exactly on a line; the take's audio and its clock
driver	A real player's clip, retimed to the song, that Motion Sync borrows motion from
gate	A check that can refuse or demand a read; never a substitute for the read
read	A recorded look at evidence at full size with text per person; the only thing that promotes to PASS
register	APPROVED_FRAMES.json, TAKES_REGISTRY.json: what has been read and what was seen
cut sheet	The JSON that IS the cut; deterministic render input
lock / master	Dan's word on an export; immutable copy with checksum and manifest; nobody else may use the words
swap chain	Fixing a wide one man per pass, naming every other person as unchanged
rule 26 look	The close-up recipe that reads real: MJ raw photo, big face, black room, one warm light, side key, expression, grain

Sources, all in the private repo dpoore12/sammy-rane-studio: HANDOFF.md, LOCKED_RULES.md, PLAYBOOK.md, handoff/engineering/SAMMY_STUDIO_SERVER_BRIEF.md, handoff/engineering/HOTEL_CALL_AUDIT_AND_EXECUTOR_PROPOSAL.md, build/*.py, handoff/*/DAN_NOTES.md. Questions to Dan; the operator on the Mac answers technical ones in writing in the repo.